COMMUNITY GUIDE

Build your own: parts, cost, and difficulty

Parts, planning costs, tested hardware, and a staged first build.

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PowerGlove Vision lets you use a camera-recognized hand to control games on RetroPie. The **PowerGlove Vision Controller (Arduino UNO Q)** handles the camera, recognition, and website. RetroPie runs the game. You do not need an original Power Glove or electronics attached to your hand.

This is a hobby project with working gameplay, not a finished consumer accessory. Native Super Glove Ball is playable, but noticeable movement latency remains. Choose it if you enjoy experimenting and can work through a little Linux setup.

What you need

Part	What to look for	Evidence and limits
PowerGlove Vision Controller	Arduino UNO Q, provisioned with Arduino App Lab	This is the project's supported Controller platform. Other Arduino UNO boards are not substitutes. This guide does not establish equivalent performance across RAM variants.
USB camera	A Linux-compatible UVC webcam with a stable mount	Razer Kiyo Pro is the camera used in the recorded cabinet tests. Other UVC cameras need validation; sharing a USB connector does not guarantee the same formats or timing.
Powered USB hub or dock	USB data connectivity for the camera and compatible Controller power arrangement	A powered hub is part of the documented setup. Ethernet through a USB dock is used on the project cabinet; a specific dock model has not been recorded as a universal recommendation.
Power supplies and data cables	Supplies appropriate to the Controller, dock, and RetroPie computer	Check power delivery and upstream data roles before buying. A charge-only cable will not connect a camera.
RetroPie computer	An already working RetroPie installation, storage, and display	Verify ordinary gamepad play first. Native Super Glove Ball additionally needs the separate custom Nestopia core built on the target.
Conventional gamepad	A working USB or Bluetooth controller for the cabinet	Keep it available for RetroArch setup, Libretro setup, EmulationStation setup, menus, and comparison tests.
Browser and local network	Computer, phone, or tablet; a LAN shared by both devices	The browser controls setup. It does not perform recognition. Wi-Fi or supported USB Ethernet can connect the Controller.
Optional extras	Camera stand, front lighting, cable ties, plain glove	Start with a bare hand. Glove colour is a diagnostic label, not a different recognition model.

Arduino lists a USB-C port supporting host/device roles and a 5 V, up-to-3 A USB-C supply specification for the board. Follow the manufacturer's power and hub guidance; the camera and dock add their own requirements. See the official UNO Q specifications before choosing the power arrangement. The recorded camera model is documented by Razer.

Budget before you buy

These are **planning allowances**, not vendor quotations or a promise that a particular accessory will work. Amounts appear in Canadian dollars (CA$), euros (€), British pounds (£), and US dollars (US$), in that order. They exclude shipping, regional tax differences, games, and a display. Check local prices and returns policies.

Conversions use the [European Central Bank reference rates for September 4, 2026]: CA$1.6038 / €1 / £0.85898 / US$1.1622. Planning figures are rounded to the nearest five currency units; totals are converted and rounded independently. These are currency equivalents of the same allowance, not researched local retail prices.

Item	CA$	€	£	US$
Controller board	CA$95-135	€60-85	£50-75	US$70-100
UVC camera	CA$65-240	€40-150	£35-130	US$45-175
Powered hub/dock	CA$50-110	€30-70	£25-60	US$35-80
Controller/dock power	CA$30-55	€20-35	£15-30	US$25-40
Cables and camera mount	CA$25-55	€15-35	£15-30	US$15-40
RetroPie kit and storage, if needed	CA$160-290	€100-180	£85-155	US$115-210
Gamepad, if needed	CA$25-55	€15-35	£15-30	US$15-40

Reuse a working cabinet, gamepad, camera, cables, and mount where possible. Check hub data/power compatibility, and avoid budgeting twice for included supplies.

Budget scenario	CA$	€	£	US$
Existing cabinet: first five items	CA$265-600	€165-375	£140-320	US$190-435
Example mid-range setup	CA$330	€205	£175	US$240
Additional RetroPie kit and gamepad	CA$185-345	€115-215	£100-185	US$135-250

The example uses a board, camera, hub, power supply, and cables/mount. These estimates are not a complete cabinet build cost. Expensive cameras, displays, and enclosure work can raise the total substantially.

As a dated retail reference, Arduino's EU store showed the UNO Q 2GB at **CA$96.07 / €59.90 / £51.45 / US$69.62** on September 6, 2026, including EU VAT. Only the euro amount is the store quotation; the other amounts are conversions using the rates above, not Canadian, UK, or US offers. [Arduino store]

Difficulty and time

No soldering or glove construction is required by the documented setup. You should be comfortable opening a terminal, copying commands carefully, finding network addresses, and checking a service when a step fails.

Allow an afternoon for a first attempt with provisioned hardware and a working RetroPie system. That is a planning estimate, not a measured installation time. Downloads, board provisioning, a native-core compile, unfamiliar networking, or camera trouble can extend it. Enclosure building and installing RetroPie from scratch are separate tasks.

The easiest starting point is local **Play -> Rock Paper Scissors** and **Glove Academy**. Neither requires a paired RetroPie cabinet, so they let you verify the camera and learn the gestures before debugging game delivery.

Assemble and test in stages

1. Make sure the RetroPie system already plays a game with its conventional controller.
2. Provision the Controller using Arduino's supported App Lab workflow. Arrange its power, powered hub/dock, and camera according to the hardware guidance.
3. Put both machines on the same reachable LAN. A USB Ethernet link can be used alongside Wi-Fi; the addresses may differ. Record their hostnames and current addresses.
4. Follow the Installation Guide for matching software on both machines. Use its maintained commands rather than copying an old release command from a forum post.
5. Open Glove Academy, choose your player, and verify that the whole hand stays visible. Work through the sixteen lessons and save your centre.
6. Pair through secure Setup. Both code and password pairing require the Controller's matrix PIN and certificate-ID check; the RetroPie code or SSH password is additional.
7. Start with a standard game mapping. Try native Super Glove Ball after ordinary delivery works, using the Gameplay Guide.
8. Export each player's hand setup to your browser device and keep a named backup. See where the files live.

Keep the camera and resting position fixed during testing. A green Networking indicator means a physical link is up; the two console checks establish separate service and authentication results. None proves the game received an input.

Before calling the build finished

Check clean starts, a stable hand view, intentional Start/Select gestures, ordinary gamepad fallback, and a short game session. A Glove Master award demonstrates lesson completion, not measured cabinet latency. Use the symptom guide when a stage fails and change one thing at a time.

To help another hobbyist reproduce your result, record the board variant, camera and dock model, OS/core versions, display settings, lighting, and what you actually tested. Keep pairing tokens and private camera footage out of public reports. The project distributes no ROM images.